SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1708**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Lokesh S (192511037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Lokesh S

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks
1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks
1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

QUESTION 1 – RAIN AND WET GROUND

Question

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Show the Empty Clause (□) if the goal is proved.

Answer

1. Representation of the Statements in Propositional Logic

To represent the given knowledge base using propositional logic, we assign a propositional symbol to each basic statement.

Let:

R = It is raining.

W = The ground is wet.

The first statement in the knowledge base is:

If it rains, then the ground becomes wet.

This can be represented as the implication:

$R \rightarrow W$

The second statement is:

It is raining.

Therefore, it is represented as:

R

The goal that we need to prove is:

W

Thus, the complete knowledge base can be represented as:

$R \rightarrow W$

R

Propositional Representation Table

Propositional Symbol	Meaning
R	It is raining
W	The ground is wet
$R \rightarrow W$	If it rains, then the ground becomes wet

Therefore, the problem can be expressed as:

Given: $R \rightarrow W$ and R

To prove: W

2. Conversion of the Statements into CNF

Resolution works on clauses in **Conjunctive Normal Form (CNF)**.

A logical expression is in CNF when it is represented as a conjunction of one or more clauses, where each clause contains literals connected by OR (∨).

The important implication conversion rule is:

$$P \rightarrow Q \equiv \neg P \lor Q$$

Applying this rule to the first statement:

$$R \rightarrow W$$

we get:

$$\neg R \lor W$$

Therefore, the first statement in CNF is:

$$\neg R \lor W$$

The second statement is simply:

R

Since R is already a single literal, it is already in CNF form.

Therefore, the clauses obtained from the knowledge base are:

$$C_1 = \neg R \lor W$$

$$C_2 = R$$

These clauses represent the complete knowledge base in CNF.

CNF Clauses

Clause	CNF Representation	Source
C_1	$\neg R \lor W$	$R \rightarrow W$
C_2	R	Given fact

3. Negation of the Goal

The goal that we need to prove is:

W

In the Resolution Algorithm, we normally prove a statement using **resolution by refutation**.

In resolution by refutation, instead of directly proving the goal, we assume that the goal is false.

Therefore, we negate the goal.

Original goal:

W

Negation of the goal:

¬W

This negated goal is then added to the existing CNF clauses.

Therefore, our complete set of clauses becomes:

C_1 = ¬R ∨ W

C_2 = R

C_3 = ¬W

where:

- C_1 is obtained from the implication.
- C_2 is the given fact that it is raining.
- C_3 is the negation of the goal.

Complete Clause Set

Clause	Expression	Purpose
C_1	¬R ∨ W	Converted implication
C_2	R	Given fact
C_3	¬W	Negated goal

The purpose of adding **¬W** is to determine whether the assumption that "the ground is not wet" leads to a contradiction.

If a contradiction is obtained, resolution produces the **Empty Clause (□)**.

4. Apply the Resolution Algorithm

The basic resolution rule can be expressed as:

If we have:

A ∨ B

and

¬A ∨ C

then resolution allows us to derive:

B ∨ C

In this problem, we have:

$\mathbf{C_1 = \neg R \lor W}$

and

$\mathbf{C_2 = R}$

The literals **¬R** and **R** are complementary literals.

Therefore, they can be eliminated through resolution.

Resolution Step 1

Take:

$\mathbf{C_1 = \neg R \lor W}$

and:

$\mathbf{C_2 = R}$

The complementary literals are:

¬R and R

After resolving them, the remaining literal is:

W

Therefore:

$\mathbf{(\neg R \lor W), R \Rightarrow W}$

Hence, we obtain a new clause:

$\mathbf{C_4 = W}$

This means that from the knowledge base, we have derived that the ground is wet.

5. Resolution with the Negated Goal

We now have:

$\mathbf{C_4 = W}$

The negated goal that we introduced earlier is:

$\mathbf{C_3 = \neg W}$

Therefore, we resolve:

$\mathbf{C_4 = W}$

with:

$\mathbf{C_3 = \neg W}$

Here, **W** and **¬W** are complementary literals.

Therefore:

$\mathbf{W, \neg W \Rightarrow} \square$

The result is:

□

This symbol is called the **Empty Clause**.

6. Complete Resolution Derivation

The complete resolution process can be written as follows:

Initial Clauses

$\mathbf{C_1 = \neg R \lor W}$

$\mathbf{C_2 = R}$

$\mathbf{C_3 = \neg W}$

First Resolution

Resolve C_1 and C_2:

$\mathbf{(\neg R \lor W) + R}$

Therefore:

$\mathbf{C_4 = W}$

Second Resolution

Resolve C_4 and C_3:

$\mathbf{W + \neg W}$

Therefore:

□

So the complete derivation is:

$\mathbf{C_1 + C_2 \rightarrow C_4}$

$\mathbf{C_4 + C_3 \rightarrow \square}$

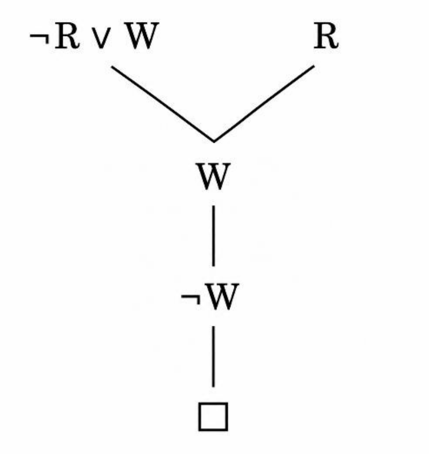
7. Resolution Table

Step	Clauses Used	Complementary Literals	Result
1	$C_1 = \neg R \lor W$, $C_2 = R$	$\neg R$ and R	$C_4 = W$
2	$C_4 = W$, $C_3 = \neg W$	W and $\neg W$	□

8. Resolution Tree

The resolution process can also be represented using a resolution tree:

$\neg R \lor W$ R



The first resolution produces **W**.

The second resolution combines **W** with **¬W** and produces the Empty Clause □.

9. Explanation of the Empty Clause

The Empty Clause □ represents a contradiction.

We initially assumed the negation of the goal:

¬W

which means:

The ground is not wet.

However, the knowledge base contains:

R

and:

R → W

From these statements, resolution derives:

W

Therefore, we have both:

W

and:

¬W

These two statements cannot simultaneously be true.

Their resolution produces:

□

Therefore, the assumption **¬W** is false.

Since the negation of the goal is false, the original goal must be true.

10. Final Conclusion

The knowledge base states that it is raining and that if it rains, the ground becomes wet.

After converting the knowledge base into CNF and adding the negation of the goal, the Resolution Algorithm produces the following sequence:

$(\neg R \lor W) + R \rightarrow W$

$W + \neg W \rightarrow \square$

Since the **Empty Clause (□)** is derived, the negation of the goal leads to a contradiction.

Therefore, the original goal is logically valid.

Hence, it is proved using the Resolution Algorithm that the ground is wet.

QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION

Question

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm.
5. Conclude whether the goal is proved.

Answer

1. Propositional Logic Representation

To represent the given knowledge base using propositional logic, suitable symbols are assigned to the statements.

Let:

S = Rahul submitted the assignment.

M = Rahul receives internal marks.

The first statement says:

If a student submits the assignment, then the student receives internal marks.

This is an implication and can be represented as:

$S \rightarrow M$

The second statement says:

Rahul submitted the assignment.

Therefore:

S

The goal that has to be proved is:

M

Therefore, the complete knowledge base can be written as:

S → M

S

Goal: M

Propositional Representation

Symbol	Meaning
S	Rahul submitted the assignment
M	Rahul receives internal marks
S → M	If Rahul submits the assignment, Rahul receives internal marks

The logical relationship can therefore be represented as:

S → M, S ⊨ M

This means that the goal **M** should logically follow from the given knowledge base.

2. Conversion of the Premises into CNF

The Resolution Algorithm requires the knowledge base to be represented in **Conjunctive Normal Form (CNF)**.

The standard rule for converting an implication into CNF is:

P → Q ≡ ¬P ∨ Q

Applying this rule to:

S → M

we obtain:

¬S ∨ M

Therefore, the first premise becomes:

¬S ∨ M

The second premise:

S

is already a single propositional literal, so it is already in CNF form.

Therefore, the CNF clauses obtained from the knowledge base are:

C_1 = ¬S ∨ M

C_2 = S

CNF Clause Representation

Clause	CNF Expression	Source
C_1	$\neg S \lor M$	From $S \rightarrow M$
C_2	S	Given fact

These two clauses represent the original knowledge base in CNF.

3. Negation of the Goal

The goal is:

M

In resolution by refutation, the goal is proved by assuming that the goal is false.

Therefore, the goal is negated:

¬M

This negated goal is added to the existing set of clauses.

Thus, we obtain:

$C_1 = \neg S \lor M$

C_2 = S

$C_3 = \neg M$

The purpose of adding **¬M** is to determine whether the assumption that Rahul does not receive internal marks creates a contradiction.

If the resolution process produces the **Empty Clause (□)**, then the negated goal is inconsistent with the knowledge base. Therefore, the original goal is proved.

Complete Clause Set

Clause	Expression	Purpose
C_1	$\neg S \lor M$	Converted implication
C_2	S	Given fact
C_3	$\neg M$	Negated goal

4. Apply the Resolution Algorithm

The resolution rule is used to combine clauses containing complementary literals.

For example:

$A \lor B$

and

¬A

can be resolved to obtain:

B

In this problem, the first resolution involves the complementary literals:

$\neg S$ and S

The second resolution involves:

M and $\neg M$

Resolution Step 1

Consider the first two clauses:

$C_1 = \neg S \lor M$

$C_2 = S$

The literals **$\neg S$** and **S** are complementary.

Therefore, they can be eliminated through resolution.

Starting with:

$(\neg S \lor M) + S$

the complementary pair **$\neg S$** and **S** is removed.

The remaining literal is:

M

Therefore:

$C_1 + C_2 \rightarrow C_4$

where:

$C_4 = M$

Resolution Expression

$(\neg S \lor M), S \Longrightarrow M$

This means that from the fact that Rahul submitted the assignment and the rule connecting assignment submission to internal marks, we derive:

M = Rahul receives internal marks.

5. Resolution with the Negated Goal

We have now derived:

$C_4 = M$

However, we also have the negated goal:

$C_3 = \neg M$

Therefore, the next resolution step is between:

$C_4 = M$

and:

$C_3 = \neg M$

The literals **M** and **¬M** are complementary.

Therefore, resolving these two clauses gives:

$M + \neg M \Rightarrow \square$

The resulting clause is:

$\square$

The symbol $\square$ represents the **Empty Clause**.

6. Complete Resolution Derivation

The entire resolution process can be summarized as follows.

Initial Clauses

$C_1 = \neg S \lor M$

$C_2 = S$

$C_3 = \neg M$

First Resolution

Resolve **C_1** and **C_2**:

$(\neg S \lor M) + S \Rightarrow M$

Therefore:

$C_4 = M$

Second Resolution

Resolve **C_4** and **C_3**:

$M + \neg M \Rightarrow \square$

Therefore:

$\square$ = Empty Clause

Resolution Table

Step	Clauses Resolved	Complementary Literals	Result
1	$C_1 = \neg S \lor M$ and $C_2 = S$	$\neg S$, S	$C_4 = M$
2	$C_4 = M$ and $C_3 = \neg M$	M, $\neg M$	$\square$

7. Explanation of Each Resolution Step

Step 1: Deriving Internal Marks

The first clause:

¬S ∨ M

represents the rule:

If Rahul submits the assignment, then Rahul receives internal marks.

The second clause:

S

states that Rahul actually submitted the assignment.

Since **S** is true, the condition of the rule is satisfied.

Resolution therefore derives:

M

which means:

Rahul receives internal marks.

Step 2: Producing the Empty Clause

The negated goal was:

¬M

This temporarily assumes:

Rahul does not receive internal marks.

But the first resolution step already derived:

M

Therefore, we have both:

M

and:

¬M

These are contradictory statements.

Resolution between these complementary literals produces:

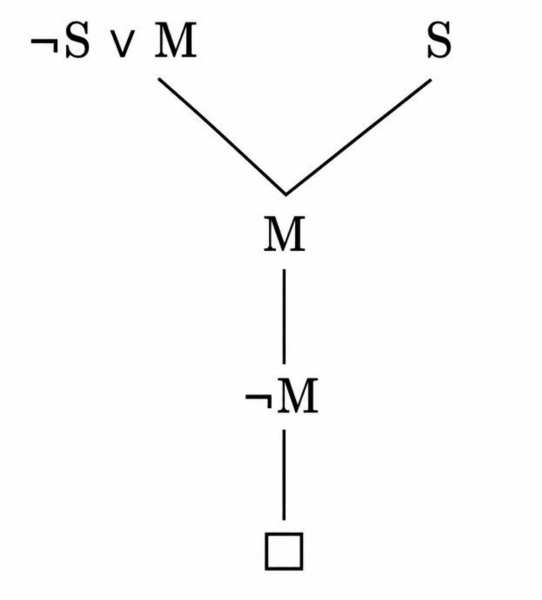
□

Therefore:

M + ¬M ⟹ □

8. Resolution Tree

The resolution tree can be represented as:



The tree shows the complete reasoning:

¬S ∨ M + S → M

and then:

M + ¬M → □

9. Interpretation of the Empty Clause

The Empty Clause □ is an important result in the Resolution Algorithm.

It represents a contradiction.

In this problem, we initially assumed:

¬M

which means:

Rahul does not receive internal marks.

However, the knowledge base itself allows us to derive:

M

which means:

Rahul receives internal marks.

Therefore, the assumption **¬M** cannot be true together with the given knowledge base.

The contradiction is represented by:

□

Thus, the negation of the goal is rejected.

10. Final Conclusion

The given knowledge base contains the rule:

S → M

and the fact:

S

After converting the rule into CNF, negating the goal, and applying the Resolution Algorithm, we obtain:

$$(\neg S \lor M) + S \Rightarrow M$$

Then:

$$M + \neg M \Rightarrow \square$$

Since the **Empty Clause (□)** has been derived, the negation of the goal leads to a contradiction. Therefore, the original goal is logically valid.

Hence, Rahul receives internal marks is proved using the Resolution Algorithm.

QUESTION 3 – LIBRARY MEMBERSHIP

Question

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm.
5. Conclude whether the goal is proved.

Answer

1. Propositional Logic Representation

To solve the problem using the Resolution Algorithm, the statements in the knowledge base are first represented using propositional logic.

Let:

L = Priya is a library member.

B = Priya can borrow books.

The first statement is:

If a person is a library member, then the person can borrow books.

This is represented as an implication:

L → B

The second statement is:

Priya is a library member.

Therefore, it is represented as:

L

The goal that needs to be proved is:

B

Therefore, the complete logical representation of the problem is:

L → B

L

Goal: B

Propositional Representation

Symbol	Meaning
L	Priya is a library member
B	Priya can borrow books
L → B	If Priya is a library member, then Priya can borrow books

Thus, the knowledge base states that Priya is a library member and that library members can borrow books.

The objective is to prove:

B

2. Conversion of the Premises into CNF

The Resolution Algorithm operates on clauses in **Conjunctive Normal Form (CNF)**.

The implication in the knowledge base is:

L → B

The standard implication conversion rule is:

P → Q ≡ ¬P ∨ Q

Applying this rule:

L → B ≡ ¬L ∨ B

Therefore, the first premise is converted into:

¬L ∨ B

The second premise is:

L

Since it is already a single propositional literal, it is already in CNF form.

Therefore, the two clauses obtained from the knowledge base are:

$C_1 = \neg L \lor B$

$C_2 = L$

CNF Clauses

Clause	CNF Expression	Source
C_1	$\neg L \lor B$	From the rule $L \rightarrow B$
C_2	L	Given fact

Therefore, the knowledge base in CNF is:

$(\neg L \lor B) \land L$

3. Negation of the Goal

The goal of the problem is:

B

which means:

Priya can borrow books.

To prove the goal using resolution by refutation, the goal is temporarily assumed to be false.

Therefore, the negation of the goal is:

$\neg B$

This negated goal is added to the existing set of clauses.

Therefore, the complete set of clauses becomes:

$C_1 = \neg L \lor B$

$C_2 = L$

$C_3 = \neg B$

Here:

- **C_1** is the CNF form of the given rule.
- **C_2** is the given fact that Priya is a library member.
- **C_3** is the negation of the goal.

Complete Clause Set

Clause	Expression	Purpose
C_1	$\neg L \lor B$	Converted implication
C_2	L	Given fact
C_3	$\neg B$	Negated goal

The purpose of introducing **$\neg B$** is to check whether assuming that Priya cannot borrow books results in a contradiction.

If a contradiction is obtained and the Empty Clause □ is derived, then the original goal **B** is proved.

4. Apply the Resolution Algorithm

The Resolution Algorithm works by identifying **complementary literals** in two clauses.

Two literals are complementary when one is the positive form and the other is its negation.

For example:

L and ¬L

or:

B and ¬B

In this problem, two resolution steps are required to obtain the Empty Clause.

Resolution Step 1

Consider the following clauses:

$C_1 = \neg L \lor B$

$C_2 = L$

The complementary literals are:

¬L and L

Therefore, we can resolve the two clauses.

Starting with:

$(\neg L \lor B) + L$

the complementary literals **¬L** and **L** are eliminated.

The remaining literal is:

B

Therefore:

$(\neg L \lor B), L \Rightarrow B$

Hence, we obtain a new clause:

$C_4 = B$

This means that the knowledge base allows us to derive the statement:

Priya can borrow books.

5. Resolution with the Negated Goal

After the first resolution step, we have derived:

$C_4 = B$

However, we also have the negated goal:

$C_3 = \neg B$

Therefore, the next resolution step uses:

$C_4 = B$

and:

$C_3 = \neg B$

The literals **B** and **¬B** are complementary.

Therefore:

$B + \neg B \Longrightarrow \square$

The result is:

$\square$

where $\square$ represents the **Empty Clause**.

6. Complete Resolution Derivation

The complete set of initial clauses is:

$C_1 = \neg L \lor B$

$C_2 = L$

$C_3 = \neg B$

First Resolution

Resolve C_1 and C_2:

$(\neg L \lor B) + L \Longrightarrow B$

Therefore:

$C_4 = B$

Second Resolution

Resolve C_4 and C_3:

$B + \neg B \Longrightarrow \square$

Therefore:

$\square$ = Empty Clause

The complete resolution sequence is:

$C_1 + C_2 \rightarrow C_4 = B$

$C_4 + C_3 \rightarrow \square$

7. Resolution Table

Step	Clauses Used	Complementary Literals	Result
1	$C_1 = \neg L \lor B$ and $C_2 = L$	¬L and L	$C_4 = B$
2	$C_4 = B$ and $C_3 = \neg B$	B and ¬B	$\square$

8. Detailed Explanation of Resolution

First Resolution

The first clause:

¬L ∨ B

represents the rule:

If Priya is a library member, then Priya can borrow books.

The second clause:

L

represents the fact:

Priya is a library member.

Since **L** is true and the rule contains **¬L ∨ B**, resolution eliminates the complementary pair **¬L** and **L**.

This leaves:

B

Therefore:

B = Priya can borrow books

has been derived.

Second Resolution

The negated goal was:

¬B

This represents the temporary assumption:

Priya cannot borrow books.

However, the first resolution step has already derived:

B

which means:

Priya can borrow books.

Therefore, we now have both:

B

and:

¬B

These two literals contradict each other.

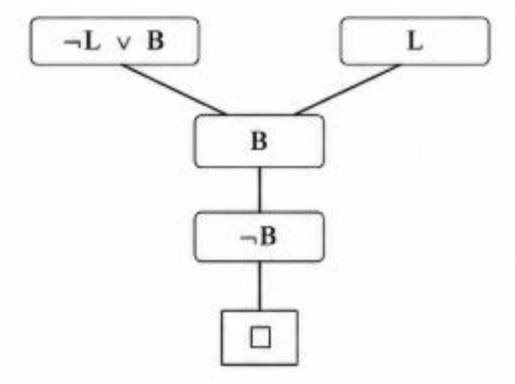
Resolution between these complementary literals gives:

B + ¬B ⟹ □

The Empty Clause indicates that the assumption **¬B** is inconsistent with the knowledge base.

9. Resolution Tree

For the resolution proof, the derivation can be represented as:



The first resolution derives **B**.

The second resolution combines **B** with **¬B** and produces the Empty Clause □.

10. Interpretation of the Empty Clause

The Empty Clause □ represents a contradiction in the set of clauses.

In this problem, the contradiction occurs because:

B

was derived from the knowledge base, while:

¬B

was introduced as the negation of the goal.

Since both cannot be true simultaneously, the assumption that:

Priya cannot borrow books

must be false.

Therefore, the original goal:

Priya can borrow books

must be true.

The derivation of the Empty Clause is therefore sufficient to establish the validity of the goal.

11. Final Conclusion

The given knowledge base contains the rule:

L → B

and the fact:

L

After converting the implication into CNF, we obtain:

C_1 = ¬L ∨ B

$C_2 = L$

The goal is **B**, so its negation:

$C_3 = \neg B$

is added.

Applying the Resolution Algorithm:

$(\neg L \lor B) + L \Rightarrow B$

Then:

$B + \neg B \Rightarrow \square$

Since the **Empty Clause (□)** is derived, the negation of the goal leads to a contradiction.

Therefore, the original goal is logically valid.

Hence, it is proved using the Resolution Algorithm that Priya can borrow books.

QUESTION 4 – PLACEMENT ELIGIBILITY

Question

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert the statements into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Show the Empty Clause (□) if the goal is proved.

Answer

1. Representation of the Statements in Propositional Logic

To solve the problem using the Resolution Algorithm, the statements given in the knowledge base are first represented using propositional logic.

Let:

A = Arun cleared the aptitude test.

E = Arun is eligible for placement.

The first statement is:

If a student clears the aptitude test, then the student is eligible for placement.

This represents an implication between two propositions.

Therefore:

A → E

The second statement is:

Arun cleared the aptitude test.

Therefore:

A

The goal that needs to be proved is:

E

Thus, the complete logical representation is:

A → E

A

Goal: E

Propositional Representation

Symbol	Meaning
A	Arun cleared the aptitude test
E	Arun is eligible for placement
A → E	If Arun clears the aptitude test, Arun is eligible for placement

Therefore, the problem can be expressed as:

A → E, A ⊨ E

This means that the goal **E** should logically follow from the given knowledge base.

2. Conversion of the Statements into CNF

The Resolution Algorithm requires the statements to be converted into **Conjunctive Normal Form (CNF)**.

The standard rule used to eliminate an implication is:

P → Q ≡ ¬P ∨ Q

The first statement is:

A → E

Applying the implication conversion rule:

A → E ≡ ¬A ∨ E

Therefore, the first statement in CNF is:

¬A ∨ E

The second statement is:

A

Since **A** is already a single literal, it is already in CNF form.

Therefore, the clauses obtained from the knowledge base are:

$\mathbf{C_1 = \neg A \lor E}$

$\mathbf{C_2 = A}$

CNF Clause Representation

Clause	CNF Expression	Source
C_1	$\neg A \lor E$	From $A \rightarrow E$
C_2	A	Given fact

Therefore, the knowledge base in CNF is:

$\mathbf{(\neg A \lor E) \land A}$

3. Negation of the Goal

The original goal is:

E

which means:

Arun is eligible for placement.

To prove the goal using resolution by refutation, we assume that the goal is false.

Therefore, the goal is negated:

$\mathbf{\neg E}$

This negated goal is added to the set of CNF clauses.

Therefore, the complete set of clauses becomes:

$\mathbf{C_1 = \neg A \lor E}$

$\mathbf{C_2 = A}$

$\mathbf{C_3 = \neg E}$

Here:

- $\mathbf{C_1}$ represents the placement eligibility rule.
- $\mathbf{C_2}$ represents the fact that Arun cleared the aptitude test.
- $\mathbf{C_3}$ represents the negation of the goal.

Complete Clause Set

Clause	Expression	Purpose
C_1	$\neg A \lor E$	Converted implication
C_2	A	Given fact
C_3	$\neg E$	Negated goal

The purpose of adding **¬E** is to determine whether the assumption that Arun is **not eligible for placement** creates a contradiction with the knowledge base.
If the Resolution Algorithm produces the Empty Clause □, the goal is proved.

4. Apply the Resolution Algorithm

The Resolution Algorithm works by identifying **complementary literals** between two clauses.
Two literals are complementary if one is the positive form and the other is its negation.
For example:
A and ¬A
are complementary.
Similarly:
E and ¬E
are complementary.
In this problem, two resolution steps are sufficient to derive the Empty Clause.

Resolution Step 1

Consider the following clauses:
$C_1 = \neg A \lor E$
$C_2 = A$
The complementary literals are:
¬A and A
Therefore, these literals can be eliminated using the resolution rule.
Starting with:
$(\neg A \lor E) + A$
the complementary pair **¬A** and **A** is removed.
The remaining literal is:
E
Therefore:
$(\neg A \lor E), A \Rightarrow E$
Hence, a new clause is obtained:
$C_4 = E$
This means that from the knowledge base, we have derived:
Arun is eligible for placement.

5. Resolution with the Negated Goal

We have obtained:

$\mathbf{C_4 = E}$

The negated goal introduced earlier is:

$\mathbf{C_3 = \neg E}$

Therefore, the next resolution step uses:

$\mathbf{C_4 = E}$

and:

$\mathbf{C_3 = \neg E}$

The literals **E** and **¬E** are complementary.

Therefore:

$\mathbf{E + \neg E \Longrightarrow \square}$

The resulting clause is:

□

The symbol □ represents the **Empty Clause**.

6. Complete Resolution Derivation

The initial clauses are:

$\mathbf{C_1 = \neg A \lor E}$

$\mathbf{C_2 = A}$

$\mathbf{C_3 = \neg E}$

First Resolution

Resolve $\mathbf{C_1}$ and $\mathbf{C_2}$:

$\mathbf{(\neg A \lor E) + A \Longrightarrow E}$

Therefore:

$\mathbf{C_4 = E}$

Second Resolution

Resolve $\mathbf{C_4}$ and $\mathbf{C_3}$:

$\mathbf{E + \neg E \Longrightarrow \square}$

Therefore:

□ = **Empty Clause**

The complete resolution sequence is:

$\mathbf{C_1 + C_2 \rightarrow C_4 = E}$

$\mathbf{C_4 + C_3 \rightarrow \square}$

7. Resolution Table

Step	Clauses Used	Complementary Literals	Result
1	$C_1 = \neg A \lor E$ and $C_2 = A$	$\neg A$ and A	$C_4 = E$
2	$C_4 = E$ and $C_3 = \neg E$	E and $\neg E$	□

8. Detailed Explanation of Resolution

Step 1 – Using the Aptitude Test Result

The first clause is:

$\neg A \lor E$

This represents the rule:

If Arun clears the aptitude test, then Arun is eligible for placement.

The second clause is:

A

This represents the fact:

Arun cleared the aptitude test.

Since **A** and **¬A** are complementary literals, they are resolved.

The remaining literal is:

E

Therefore:

$C_1 + C_2 \Rightarrow E$

This gives us:

E = Arun is eligible for placement

Step 2 – Contradiction with the Negated Goal

The negated goal was:

¬E

which temporarily assumes:

Arun is not eligible for placement.

However, the first resolution step already derived:

E

which states:

Arun is eligible for placement.

Therefore, we now have:

E

and:

¬E

These are contradictory.

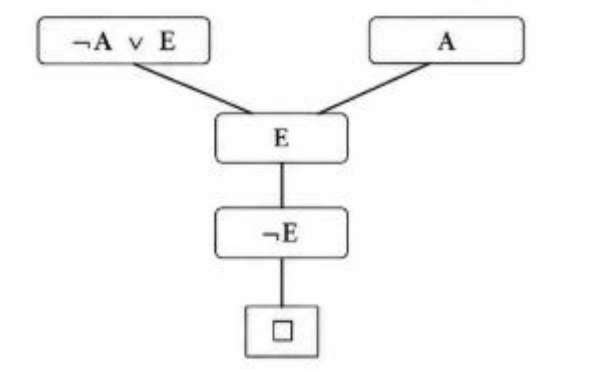
Resolving the two complementary literals gives:

E + ¬E ⟹ □

Thus, the Empty Clause is obtained.

9. Resolution Tree

The resolution proof can be represented as:



The first resolution produces **E**.

The second resolution combines **E** with **¬E** and produces the Empty Clause □.

10. Explanation of the Empty Clause

The Empty Clause □ represents a contradiction.

Initially, we assumed the negation of the goal:

¬E

This means:

Arun is not eligible for placement.

However, the knowledge base contains:

A

which states that Arun cleared the aptitude test.

It also contains the rule:

A → E

which means that clearing the aptitude test leads to placement eligibility.

After converting the rule into CNF:

¬A ∨ E

resolution with **A** produces:

E

Therefore, the knowledge base establishes **E**, while our temporary assumption establishes **¬E**.

Since both cannot be true at the same time, the contradiction produces:

□

This proves that the assumption **¬E** is false.

11. Interpretation of the Result

The resolution process can be understood logically as follows:

1. Arun cleared the aptitude test.
2. Students who clear the aptitude test are eligible for placement.
3. Therefore, Arun is eligible for placement.
4. Assuming that Arun is not eligible produces a contradiction.
5. The contradiction is represented by the Empty Clause □.

Thus, the Resolution Algorithm successfully proves the required goal.

12. Final Conclusion

The given knowledge base contains:

A → E

and:

A

After converting the implication into CNF:

A → E ≡ ¬A ∨ E

The negation of the goal is:

¬E

The complete resolution process is:

(¬A ∨ E) + A ⟹ E

Then:

E + ¬E ⟹ □

Since the **Empty Clause (□)** is derived, the negation of the goal leads to a contradiction.

Therefore, the original goal is logically valid.

Hence, it is proved using the Resolution Algorithm that Arun is eligible for placement.

QUESTION 5 – ACCESS CONTROL SYSTEM

Question

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

Answer

1. Representation of the Knowledge Base in Propositional Logic

To apply the Resolution Algorithm, the statements in the knowledge base are first represented using propositional variables.

Let:

P = The user entered the correct password.

A = The user is authenticated.

G = The user is granted access.

The first statement is:

If a user enters the correct password, then the user is authenticated.

This can be represented as:

$P \rightarrow A$

The second statement is:

If a user is authenticated, then the user is granted access.

This can be represented as:

$A \rightarrow G$

The third statement is:

The user entered the correct password.

Therefore:

P

The goal is:

G

which means:

The user is granted access.

Therefore, the complete knowledge base can be represented as:

P → A

A → G

P

Goal: G

2. Propositional Representation

The propositional variables used in the problem are:

Symbol	Meaning
P	The user entered the correct password
A	The user is authenticated
G	The user is granted access

The logical relationships are:

P → A

A → G

P

The required conclusion is:

G

The reasoning process is therefore expected to follow the chain:

P → A → G

Since the fact **P** is given, we should be able to derive **A**, followed by **G**.

However, to formally prove the goal using the Resolution Algorithm, we must convert the statements into CNF and use proof by refutation.

3. Conversion of All Implications into CNF

The Resolution Algorithm works with clauses in **Conjunctive Normal Form (CNF)**.

The standard implication conversion rule is:

P → Q ≡ ¬P ∨ Q

First Implication

The first rule is:

$P \rightarrow A$

Using the implication equivalence:

$P \rightarrow A \equiv \neg P \lor A$

Therefore:

$C_1 = \neg P \lor A$

Second Implication

The second rule is:

$A \rightarrow G$

Applying the same conversion:

$A \rightarrow G \equiv \neg A \lor G$

Therefore:

$C_2 = \neg A \lor G$

Third Statement

The third statement is:

P

Since P is already a single literal, it is already in CNF.

Therefore:

$C_3 = P$

CNF Clause Representation

Clause	CNF Expression	Source
C_1	$\neg P \lor A$	From $P \rightarrow A$
C_2	$\neg A \lor G$	From $A \rightarrow G$
C_3	P	Given fact

Thus, the complete knowledge base in CNF is:

$(\neg P \lor A) \land (\neg A \lor G) \land P$

4. Negation of the Goal

The original goal is:

G

which means:

The user is granted access.

In resolution by refutation, we temporarily assume that the goal is false.

Therefore, the goal is negated:

$\neg G$

This negated goal is added to the set of clauses.

Therefore:

$C_4 = \neg G$

The complete set of clauses is now:

$C_1 = \neg P \lor A$

$C_2 = \neg A \lor G$

$C_3 = P$

$C_4 = \neg G$

Complete Clause Set

Clause	Expression	Purpose
C_1	$\neg P \lor A$	Correct-password rule
C_2	$\neg A \lor G$	Authentication-to-access rule
C_3	P	Given fact
C_4	$\neg G$	Negated goal

The objective is now to use resolution to derive the Empty Clause □.

If the Empty Clause is derived, it proves that the assumption $\neg G$ is contradictory.

5. Apply the Resolution Algorithm

This problem requires multiple resolution steps because the knowledge base contains two connected rules.

The reasoning is:

Correct Password → Authentication → Access

In propositional form:

$P \rightarrow A$

and:

$A \rightarrow G$

The fact:

P

allows us to derive **A**, and then **A** allows us to derive **G**.

Finally, **G** is resolved against the negated goal $\neg G$.

6. Resolution Step 1 – Derive Authentication

The first two relevant clauses are:

$\mathbf{C_1 = \neg P \lor A}$

$\mathbf{C_3 = P}$

The complementary literals are:

¬P and P

Therefore, resolve C_1 and C_3.

Starting with:

$\mathbf{(\neg P \lor A) + P}$

The complementary literals **¬P** and **P** are eliminated.

The remaining literal is:

A

Therefore:

$\mathbf{(\neg P \lor A), P \Rightarrow A}$

Hence, a new clause is obtained:

$\mathbf{C_5 = A}$

This means:

The user is authenticated.

7. Resolution Step 2 – Derive Granted Access

Now consider:

$\mathbf{C_2 = \neg A \lor G}$

and the newly derived clause:

$\mathbf{C_5 = A}$

The complementary literals are:

¬A and A

Therefore, resolve the two clauses:

$\mathbf{(\neg A \lor G) + A}$

After eliminating the complementary pair **¬A** and **A**, the remaining literal is:

G

Therefore:

$\mathbf{(\neg A \lor G), A \Rightarrow G}$

Hence, we obtain:

$\mathbf{C_6 = G}$

This means:

The user is granted access.

8. Resolution Step 3 – Derive the Empty Clause

The original goal is:

G

and the negated goal is:

$C_4 = \neg G$

From Step 2, we have already derived:

$C_6 = G$

Therefore, resolve:

$C_6 = G$

with:

$C_4 = \neg G$

The literals **G** and **$\neg G$** are complementary.

Therefore:

$G + \neg G \Longrightarrow \square$

The result is:

$\square$

where $\square$ represents the **Empty Clause**.

9. Complete Resolution Derivation

The initial clauses are:

$C_1 = \neg P \lor A$

$C_2 = \neg A \lor G$

$C_3 = P$

$C_4 = \neg G$

First Resolution

$C_1 + C_3 \to C_5$

Therefore:

$(\neg P \lor A) + P \Longrightarrow A$

So:

$C_5 = A$

Second Resolution

$C_2 + C_5 \to C_6$

Therefore:

$(\neg A \lor G) + A \Rightarrow G$

So:

$C_6 = G$

Third Resolution

$C_6 + C_4 \rightarrow \square$

Therefore:

$G + \neg G \Rightarrow \square$

So:

$\square$ = **Empty Clause**

10. Resolution Table

Step	Clauses Used	Complementary Literals	Result
1	$C_1 = \neg P \lor A$ and $C_3 = P$	$\neg P$ and P	$C_5 = A$
2	$C_2 = \neg A \lor G$ and $C_5 = A$	$\neg A$ and A	$C_6 = G$
3	$C_6 = G$ and $C_4 = \neg G$	G and $\neg G$	$\square$

11. Detailed Explanation of Each Resolution Step

Step 1: Correct Password Leads to Authentication

The clause:

$\neg P \lor A$

represents the statement:

If the user enters the correct password, then the user is authenticated.

The given fact is:

P

which means:

The user entered the correct password.

Since **P** and **¬P** are complementary, resolution eliminates them and derives:

A

Therefore:

A = The user is authenticated

Step 2: Authentication Leads to Access

The second rule is represented by:

$\neg A \lor G$

This means:

If the user is authenticated, then the user is granted access.

From Step 1, we already obtained:

A

Resolving:

¬A ∨ G

with:

A

eliminates **¬A** and **A**.

Therefore, we obtain:

G

Thus:

G = The user is granted access

Step 3: Contradiction with the Negated Goal

To prove the goal, we introduced:

¬G

This represents the temporary assumption:

The user is not granted access.

However, the resolution process has already derived:

G

which states:

The user is granted access.

Therefore, both:

G

and:

¬G

are present.

These two literals are complementary.

Resolving them produces:

□

This is the Empty Clause.

12. Resolution Chain

The complete reasoning can be summarized as:

P → A

A → G

P

Therefore:

P ⟹ A

Then:

A ⟹ G

Finally:

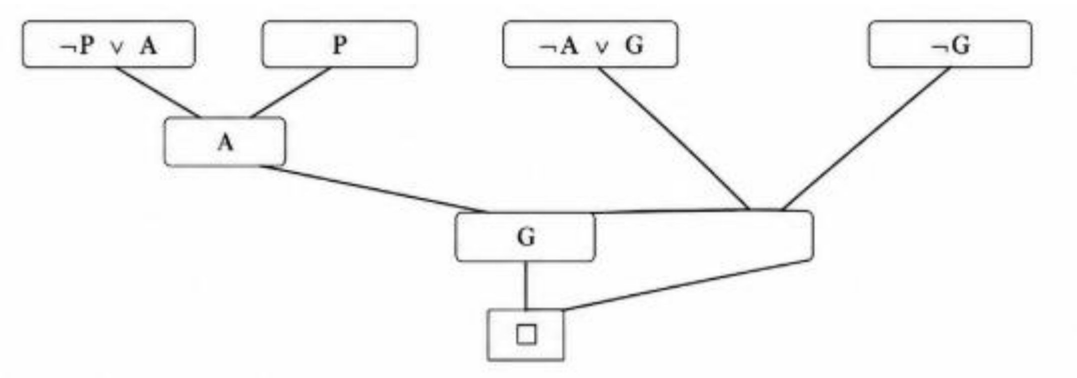
G + ¬G ⟹ □

Thus:

□ = Empty Clause

13. Resolution Tree

You can put the resolution tree separately in your Word document:



14. Interpretation of the Empty Clause

The Empty Clause □ represents a contradiction.

In this problem, we initially assumed:

¬G

which means:

The user is not granted access.

However, using the knowledge base, resolution derives:

A

and then:

G

Therefore, the assumption **¬G** contradicts the knowledge base.

The contradiction is represented by:

G + ¬G ⟹ □

Hence, the negated goal is false.

Therefore, the original goal **G** must be true.

15. Final Conclusion

The given knowledge base states that:

1. Entering the correct password makes the user authenticated.
2. An authenticated user is granted access.
3. The user entered the correct password.

After converting the implications into CNF, the clauses are:

$\mathbf{C_1 = \neg P \lor A}$

$\mathbf{C_2 = \neg A \lor G}$

$\mathbf{C_3 = P}$

The negated goal is:

$\mathbf{C_4 = \neg G}$

Applying the Resolution Algorithm:

$\mathbf{(\neg P \lor A) + P \Rightarrow A}$

$\mathbf{(\neg A \lor G) + A \Rightarrow G}$

$\mathbf{G + \neg G \Rightarrow \square}$

Since the **Empty Clause (□)** is derived, the negation of the goal leads to a contradiction.

Therefore, the original goal is logically valid.

Hence, it is proved using the Resolution Algorithm that the user is granted access.